LearnPerk Data Dictionary

Every column of every combined file, and what it means

September 2026 Generated from the data itself

How to read this

Four files, one per grain. Content views and offer impressions cannot honestly share a row, so each gets its own file, plus one long event stream that holds everything in order.

- **Values** shows *min / median / max* for numeric columns, or the list of categories for everything else. Computed from the files, so it cannot drift from what is actually there.
- **Null** is the percentage of rows where the column is empty. In this dataset a null is always meaningful: either consent said no, the event never happened, or a merchant did not disclose something. Filling nulls with zero collapses three different facts into one.
- **Columns prefixed `truth_`** are the hidden traits the simulator used to drive behaviour. They are the answer key. They are in the files so you can check whether a model recovered what was put in. **Never train on them.**

`combined_offers.csv`

One row per offer impression 9,990 rows × 92 columns

The model-ready table. Every offer the app showed, whether or not it was accepted, with the commuter, the journey, the device state and the downstream purchase all joined on.

Column	Values	Null	What it is
Keys and timing			
`impression_id`	9,990 distinct, top: o_0006451	–	Primary key. One offer shown to one commuter at one moment.
`user_id`	1,654 distinct, top: u_01289	–	The commuter this row belongs to. Joins to combined_users and to every other file.
`trip_id`	6,256 distinct, top: t_0015216	–	The journey this happened on. A trip is one or more legs with a single purpose.
`leg_id`	6,771 distinct, top: lg_0011477	–	The specific leg. A leg is one vehicle, boarding to alighting.
`shown_at`	5,428 distinct, top: 2026-08-28 08:18:00	–	When the item or offer was surfaced. All timestamps are local Amsterdam time.
`hour`	0.00 / 13.00 / 23.00	–	Hour of day, 0 to 23, taken from the timestamp. Derived, not stored separately.
`day_of_week`	1, 3, 0, 4, 2, 6, 5	–	Monday is 0, Sunday is 6.
`is_peak`	1, 0	–	1 if inside 07:00–09:00 or 16:00–18:00. Peak raises delay and crowding.
`is_weekend`	0, 1	–	1 on Saturday or Sunday. Travel probability drops sharply.
What happened			
`accepted`	0, 1	–	1 if the commuter took the offer. This is the target variable for the acceptance task.
`outcome`	ignored, declined, accepted, expired	–	accepted, declined, ignored or expired. Every impression carries one, so the data shows what a bad offer looks like as well as a good one.
The offer			
`merchant_id`	33 distinct, top: m_003	–	The merchant behind the offer. Joins to the merchants reference table.

Column	Values	Null	What it is
merchant_category	15 distinct, top: Coffee & Food-to-go	–	Display category of the merchant, such as Coffee & Food-to-go.
mcc	4,814 / 5,812 / 7,999	–	Real four-digit ISO 18245 merchant category code. Because OVpay runs on Mastercard rails, this is literally the code a real transaction would carry, even though the brand is fictional.
position	1, 2, 3	–	Rank in the list shown. Position 1 is the top slot.
discount_pct	10.0, 5.0, 15.0, 20.0, 25.0, 30.0	–	Discount offered, 5 to 30 percent. A merchant-side decision, independent of the commuter.
detour_minutes	0.00 / 3.30 / 40.00	–	How far out of their way the offer would take them, measured against the route they are already on. A partner in a zone they are passing through costs a short walk; one off the route costs the ride out and back. Suppresses acceptance, and weakens the pull of the discount as it grows. This is the main channel through which where somebody lives changes what they buy.
expires_at	6,980 distinct, top: 2026-09-04T18:42	–	When the offer stops being valid. Drives the deadline spike in redemption timing.
expiry_window_hours	2.0, 6.0, 24.0	–	How long the offer was live: 2, 6 or 24 hours.
tier_at_impression	Bronze, Silver, Gold, Platinum	–	Lifetime loyalty status: Bronze, Silver, Gold or Platinum. Earned cumulatively and never spent.
tier_progress	0.00 / 0.38 / 1.00	–	Progress from 0 to 1 toward affording the next 500-point reward. The goal-gradient term reads this: acceptance rises as it approaches 1.
propensity	0.030303	–	The probability this particular offer was chosen to be shown. Needed for any later off-policy evaluation of a ranking change; without it that analysis is impossible.
decided_at	3,906 distinct, top: 2026-09-04T08:17	42%	When the commuter accepted or declined. Blank when the offer was simply ignored.
truth_p_accept	0.00 / 0.11 / 0.92	–	The exact probability the generator used before flipping the coin. This is the answer key for the acceptance task and sets the theoretical ceiling any model can reach.
Point-in-time history (computed from earlier rows only)			
offers_seen_before	0.00 / 3.00 / 27.00	–	How many offers this commuter had already been shown, counting only impressions strictly earlier than this one.
offers_accepted_before	11 distinct, top: 0	–	How many they had already accepted, again strictly earlier.
accept_rate_before	0.00 / 0.00 / 1.00	17%	The two columns above divided. Null on a commuter's first impression, because a rate over zero offers is undefined rather than zero.
merchant_seen_before	0, 1, 2, 3, 4, 5	–	How many times this commuter had already been shown this specific merchant. Drives the repeat-merchant effect.
Who the commuter is			
birth_year	1,937 / 1,987 / 2,010	–	Year of birth. Age is always derived from this so a longer simulation window does not silently age the population.
age_at_start	16.00 / 39.00 / 89.00	–	Age on the first day of the window.
age_cell	11 distinct, top: A3	–	One of eleven bands, A1 to A11. The grid is fine enough that every published band from CBS, Reuters and the app-engagement studies can be rebuilt by adding adjacent cells.
gender	Female, Male, Prefer not to say, Not collected (mino...	–	Female, Male or Prefer not to say. Not collected for under-18s. Carries a coefficient of exactly zero on offer acceptance, because three published studies disagree on the direction.
occupation	Established profess..., Early-career profes..., Student, Retired, S...	–	Life stage, sampled conditional on age so a 19-year-old retiree never appears.
household_type	Couple no kids, Shared housing, Family with children, Solo	–	Solo, shared housing, couple or family with children.
traveler_archetype	Habitual commuter, Vacation/leisure, Occasional/personal, Busin...	–	Habitual commuter, occasional/personal, vacation/leisure or business traveller. Roughly one in ten is a non-commuter.
signup_date	643 distinct, top: 2022-10-24	–	When the account was created. Tenure is always derived from this, never stored as a number, because a stored day count is wrong tomorrow.

Column	Values	Null	What it is
device_performance_class	Mid-tier, Low-tier, High-tier	–	Low, mid or high tier, split 30/45/25 on Android market data. Low-tier devices have their cache evicted more often.
home_zone	10 distinct, top: Nieuw-West	–	Coarse neighbourhood, never an address. Drawn from the resident population of each Amsterdam zone, tilted by age: students towards Centrum, Oost and West, families towards Nieuw-West and IJburg, the over-55s towards Buitenveldert.
work_zone	10 distinct, top: Zuid	–	Coarse neighbourhood for the usual destination. Drawn conditional on home_zone from a gravity model balanced against the jobs in each zone, so somebody living in Buitenveldert is far likelier to work on the Zuidas than somebody living in Noord. Equal to home_zone for the retired and for home carers, who have no commute.
Hidden traits — the answer key, never a feature			
truth_taste_vector	1,654 distinct, top: 0.1018;0.0748;0.244...	–	Ten weights summing to 1, one per content category, stored semicolon-separated because CSV has no array type. Decided which items this commuter was shown.
truth_price_sensitivity	0.02 / 0.46 / 0.99	–	How much a price puts them off. Negative coefficient on acceptance.
truth_coupon_proneness	0.00 / 0.26 / 0.82	–	General appetite for offers. Positive coefficient. Deliberately has no age or gender shift.
truth_detour_tolerance_min	0.50 / 4.85 / 22.00	–	Minutes they will walk out of their way. Enters as a threshold rather than a slope, which is why it is the hardest trait to recover.
truth_novelty_seeking	0.02 / 0.46 / 0.98	–	Probability of being shown a new brand rather than one they already use.
truth_battery_anxiety	0.01 / 0.36 / 0.93	–	Sets a personal disengagement threshold between 20 and 40 percent battery, matching the two curves in the low-battery-anxiety study.
truth_content_appetite	0.24 / 1.00 / 3.85	–	Multiplier on how many items they open per leg.
accessibility_flag	none, reduced_mobility, visual_preference, hearing_impairment, ...	–	Opt-in. Changes how an offer or notification should be rendered, not whether it is shown.
consent_status	Opted-in, Withdrawn	–	Opted-in, Opted-out or Withdrawn. An opted-out commuter has no behavioural rows anywhere in the dataset, not blank ones.
The merchant			
merchant_name	33 distinct, top: SnelBite	–	Fictional brand name. No real merchant appears anywhere in the dataset.
merchant_cat	15 distinct, top: Coffee & Food-to-go	–	Duplicate of merchant_category carried in from the merchants table. Kept for the join, safe to drop.
merchant_partner_type	Chain, Online, Independent	–	Chain, Independent or Online.
merchant_channel	Brick-and-mortar, Digital-only, Both	–	Brick-and-mortar, Digital-only or Both.
merchant_zone	11 distinct, top: Online	–	Coarse location of the merchant, placed by footfall rather than spread evenly, so Centrum and the Zuidas carry the density they do in life. Reads `Online` for a digital-only partner, which has no shopfront and so is never a detour.
The journey			
purpose	Commute, Leisure, Personal errand, Business	–	Commute, personal errand, leisure or business. Set per trip, independent of the commuter's overall archetype.
n_legs	1, 2, 3	–	How many legs this trip had. Not drawn from a table: it falls out of the network. Two zones on the same metro line are one leg, two zones with nothing between them mean a change at Centraal, and a thinly served destination such as IJburg or Osdorp adds a last mile.
multimodal_flag	0, 1	–	1 if the trip crossed more than one mode, such as bike then tram.
total_minutes	6.00 / 15.40 / 62.20	–	Total in-vehicle minutes across the trip, summed over the legs actually taken.
planned_minutes	6.00 / 15.00 / 56.00	–	What the journey planner would have quoted: the zone-to-zone time plus an interchange penalty, before any delay. The gap between this and total_minutes is the day going wrong.

Column	Values	Null	What it is
origin_zone	10 distinct, top: Zuid	–	Where the journey started. A commute leaves from home in the morning and from work in the evening; anything else starts at home and is pulled towards retail and nightlife.
destination_zone	10 distinct, top: Zuid	–	Where the journey was heading. Together with origin_zone this determines the route, and therefore the line, the stops, the duration and which offers are a short detour.
line_id	22 distinct, top: M51	–	Real GVB or NS line, such as Tram 12 or M52.
mode	Metro, Tram, Bus, Rail, Micromobility	–	Tram, Metro, Bus, Rail or Micromobility.
board_stop	32 distinct, top: Centraal	–	Where they got on. A real stop, and one that is genuinely inside board_zone and on this line.
alight_stop	32 distinct, top: Centraal	–	Where they got off. Rail legs out of Zuid or Sloterdijk can end at Schiphol, which is the one destination modelled outside the city.
board_zone	10 distinct, top: Centrum	–	The zone this leg started in.
alight_zone	10 distinct, top: Centrum	–	The zone this leg ended in. On a multi-leg trip this is the interchange, not the final destination. A validation check refuses to write the file unless every line really serves both of its zones.
board_stop_type	Origin, Mid-route	–	Origin or Mid-route. Most people board mid-route, which leaves a shorter window to show them anything.
alight_stop_type	Destination, Mid-route	–	Destination or Mid-route.
delay_minutes	-1.00 / 0.80 / 45.00	–	Actual minus scheduled departure. Log-normal, with a 1.55x multiplier at peak.
crowding_level	Low, Medium, High	25%	Low, Medium or High. Deliberately blank where GVB would not publish occupancy.
duration_minutes	4.60 / 11.90 / 48.10	–	How long this leg took. Derived from the zone-to-zone time it covers, jittered for the walk to the stop, so a Noord-to-Centrum hop under the IJ is short and Nieuw-West to Zuidoost is long.
Device and context			
battery_pct	2 / 48 / 100	–	Charge at that moment. Decays linearly through the day at a rate inside the measured 1.06 to 5.02 percent per hour range.
connectivity_state	Online, Degraded, Offline	–	Online, Degraded or Offline. Drops on underground metro legs. No content is shown when offline.
motion_state	In-vehicle, Walking	–	In-vehicle or Walking.
screen_state	Active, Locked, Background	–	Active, Locked or Background.
app_foreground_state	Background, Foreground	–	Whether LearnPerk was the app in front.
ambient_location_precision	GPS-confirmed, Inferred	–	GPS-confirmed or Inferred. A confidence qualifier on anything location-dependent.
local_cache_state	fresh, empty, stale, evicted	–	fresh, stale, empty or evicted. Derived from connectivity history and device tier rather than drawn independently. This is the field that gates the sparse-history model fallback.
What happened afterwards			
purchase_id	1,554 distinct, top: p_0001505	84%	Set only when the offer converted. Null on every declined or ignored impression.
purchased_at	1,429 distinct, top: 2026-09-01T17:46	84%	When the purchase happened, if it did.
basket_value_eur	0.60 / 4.00 / 27.29	89%	Transaction value. Blank on roughly 28 percent of purchases, because merchants do not always disclose the amount. Blank means unknown, not zero.
latency_minutes	0 / 9 / 1,419	84%	Minutes from the offer being shown to the purchase. Bimodal: most act within minutes, a second group redeems just before expiry, so the mean is a poor summary and the median is the right one.
repeat_purchase_flag	0.0, 1.0	84%	1 if the commuter had bought from this merchant before.
redeemed_at	1,416 distinct, top: 2026-09-02T18:41	84%	When the reward was claimed at the merchant.
redemption_method	tap, qr, in_app_code	84%	tap, qr or in_app_code. Tap leads, matching Dutch contactless adoption.
redemption_zone	11 distinct, top: Online	84%	Coarse location where the reward was claimed.

Column	Values	Null	What it is
confirmation_status	confirmed, manual_override, sync_lag, eligibility_rejected, exp…	84%	One of six typed outcomes rather than a single failure flag: confirmed, manual_override (the scanner could not read the phone screen so the cashier keyed it), sync_lag, eligibility_rejected, expired_before_use or integration_outage. Only the first two award points.
points_earned	6 / 40 / 273	86%	Points added by this purchase, ten per euro.
points_spent	0.0, 500.0, 1000.0	86%	Points removed if claiming a reward was triggered at the same moment.
points_balance_after	0 / 228 / 1,515	86%	Running balance. Reconciles exactly for every commuter, checked on every run.
Derived at the moment of the impression			
age_at_impression	16.00 / 39.00 / 89.00	–	Age at the moment the offer was shown.
tenure_days_at_impression	1 / 169 / 1,411	–	Days on the platform at that moment. Feeds the sparse-signal versus rich-history decision.

combined_content_views.csv

One row per content view 19,334 rows × 43 columns

Every article or video opened on a journey leg, with how long it held attention and what the commuter did next.

Column	Values	Null	What it is
Keys and timing			
view_id	19,334 distinct, top: v_00000001	–	Primary key. One content item shown to one commuter at one moment.
user_id	1,622 distinct, top: u_00368	–	The commuter this row belongs to. Joins to combined_users and to every other file.
trip_id	5,928 distinct, top: t_0002718	–	The journey this happened on. A trip is one or more legs with a single purpose.
leg_id	6,428 distinct, top: lg_0008411	–	The specific leg. A leg is one vehicle, boarding to alighting.
The content item			
item_id	600 distinct, top: c_0490	–	The article or video shown. Joins to the content_items reference table.
display_category	10 distinct, top: Local News & Curren…	–	Workbook content category, such as Local News & Current Affairs.
format	video, text	–	video or text. Which one a commuter gets leans on their age.
shown_at	18,852 distinct, top: 2026-08-28 08:31:35	–	When the item or offer was surfaced. All timestamps are local Amsterdam time.
position	9 distinct, top: 1	–	Rank in the list shown. Position 1 is the top slot.
Engagement			
dwell_seconds	0 / 29 / 1,397	–	How long the item held attention. Two behaviours are mixed here: a screening glance of under two seconds, and a genuine read that follows a log-normal.
completion_state	started, partial, completed	–	started, partial or completed. Derived from dwell against the item's own length, so it can never contradict the dwell time.
bounced	0, 1	–	1 if the item was screened out almost immediately.
iab_tier1	News, Travel, Food & Drink, Pop Culture, Careers, Sports, Home …	–	The matching top-level group in the IAB Content Taxonomy 3.1, so the category can be looked up against third-party audience data.
length_units	20 / 145 / 2,027	–	Word count for text, seconds for video.
Reaction			
reaction	like, dismiss, not_interested, save, share	96%	like, save, share, dismiss or not_interested. Blank on most views: reactions are rare and cluster at the extremes of dwell.

Column	Values	Null	What it is
reacted_at	692 distinct, top: 2026-09-01T17:04:19	96%	When the like, save or dismiss happened. Can be well after the view.
Who the commuter is			
birth_year	1,937 / 1,989 / 2,010	–	Year of birth. Age is always derived from this so a longer simulation window does not silently age the population.
age_cell	11 distinct, top: A3	–	One of eleven bands, A1 to A11. The grid is fine enough that every published band from CBS, Reuters and the app-engagement studies can be rebuilt by adding adjacent cells.
gender	Female, Male, Not collected (mino…, Prefer not to say	–	Female, Male or Prefer not to say. Not collected for under-18s. Carries a coefficient of exactly zero on offer acceptance, because three published studies disagree on the direction.
occupation	Established profess…, Early-career profes…, Student, Retired, S…	–	Life stage, sampled conditional on age so a 19-year-old retiree never appears.
household_type	Couple no kids, Shared housing, Solo, Family with children	–	Solo, shared housing, couple or family with children.
traveler_archetype	Habitual commuter, Vacation/leisure, Occasional/personal, Busin…	–	Habitual commuter, occasional/personal, vacation/leisure or business traveller. Roughly one in ten is a non-commuter.
signup_date	642 distinct, top: 2022-10-24	–	When the account was created. Tenure is always derived from this, never stored as a number, because a stored day count is wrong tomorrow.
device_performance_class	Mid-tier, Low-tier, High-tier	–	Low, mid or high tier, split 30/45/25 on Android market data. Low-tier devices have their cache evicted more often.
truth_content_appetite	0.24 / 1.16 / 3.85	–	Multiplier on how many items they open per leg.
The journey			
purpose	Commute, Leisure, Personal errand, Business	–	Commute, personal errand, leisure or business. Set per trip, independent of the commuter's overall archetype.
is_weekend	0, 1	–	1 on Saturday or Sunday. Travel probability drops sharply.
origin_zone	10 distinct, top: Nieuw-West	–	Where the journey started. A commute leaves from home in the morning and from work in the evening; anything else starts at home and is pulled towards retail and nightlife.
destination_zone	10 distinct, top: Nieuw-West	–	Where the journey was heading. Together with origin_zone this determines the route, and therefore the line, the stops, the duration and which offers are a short detour.
line_id	22 distinct, top: M50	–	Real GVB or NS line, such as Tram 12 or M52.
mode	Tram, Metro, Rail, Bus, Micromobility	–	Tram, Metro, Bus, Rail or Micromobility.
delay_minutes	-0.90 / 0.90 / 45.00	–	Actual minus scheduled departure. Log-normal, with a 1.55x multiplier at peak.
board_zone	10 distinct, top: Zuid	–	The zone this leg started in.
alight_zone	10 distinct, top: Centrum	–	The zone this leg ended in. On a multi-leg trip this is the interchange, not the final destination. A validation check refuses to write the file unless every line really serves both of its zones.
crowding_level	Low, Medium, High	25%	Low, Medium or High. Deliberately blank where GVB would not publish occupancy.
duration_minutes	4.80 / 14.70 / 48.60	–	How long this leg took. Derived from the zone-to-zone time it covers, jittered for the walk to the stop, so a Noord-to-Centrum hop under the IJ is short and Nieuw-West to Zuidoost is long.
Device and context			
battery_pct	2 / 49 / 100	–	Charge at that moment. Decays linearly through the day at a rate inside the measured 1.06 to 5.02 percent per hour range.
connectivity_state	Online, Degraded	–	Online, Degraded or Offline. Drops on underground metro legs. No content is shown when offline.
screen_state	Active, Locked, Background	–	Active, Locked or Background.

Column	Values	Null	What it is
local_cache_state	fresh, empty, stale, evicted	–	fresh, stale, empty or evicted. Derived from connectivity history and device tier rather than drawn independently. This is the field that gates the sparse-history model fallback.
Derived at the moment of the view			
age_at_view	16.00 / 37.00 / 89.00	–	Age at the moment the content was shown.
hour	0.00 / 12.00 / 23.00	–	Hour of day, 0 to 23, taken from the timestamp. Derived, not stored separately.
has_reaction	0, 1	–	1 if any reaction followed. Convenience flag over the column above.

combined_users.csv

One row per commuter 2,000 rows × 49 columns

The profile as generated, the hidden traits that drove their behaviour, and a summary of everything they did across the window.

Column	Values	Null	What it is
Identity			
user_id	2,000 distinct, top: u_00000	–	The commuter this row belongs to. Joins to combined_users and to every other file.
Who the commuter is			
birth_year	1,937 / 1,986 / 2,010	–	Year of birth. Age is always derived from this so a longer simulation window does not silently age the population.
age_at_start	16.00 / 40.00 / 89.00	–	Age on the first day of the window.
age_cell	11 distinct, top: A9	–	One of eleven bands, A1 to A11. The grid is fine enough that every published band from CBS, Reuters and the app-engagement studies can be rebuilt by adding adjacent cells.
gender	Male, Female, Not collected (mino…, Prefer not to say	–	Female, Male or Prefer not to say. Not collected for under-18s. Carries a coefficient of exactly zero on offer acceptance, because three published studies disagree on the direction.
occupation	Established profess…, Early-career profes…, Retired, Student, B…	–	Life stage, sampled conditional on age so a 19-year-old retiree never appears.
household_type	Couple no kids, Shared housing, Solo, Family with children	–	Solo, shared housing, couple or family with children.
traveler_archetype	Habitual commuter, Occasional/personal, Vacation/leisure, Busin…	–	Habitual commuter, occasional/personal, vacation/leisure or business traveller. Roughly one in ten is a non-commuter.
signup_date	710 distinct, top: 2022-10-24	–	When the account was created. Tenure is always derived from this, never stored as a number, because a stored day count is wrong tomorrow.
device_performance_class	Mid-tier, Low-tier, High-tier	–	Low, mid or high tier, split 30/45/25 on Android market data. Low-tier devices have their cache evicted more often.
home_zone	10 distinct, top: Nieuw-West	–	Coarse neighbourhood, never an address. Drawn from the resident population of each Amsterdam zone, tilted by age: students towards Centrum, Oost and West, families towards Nieuw-West and IJburg, the over-55s towards Buitenveldert.
work_zone	10 distinct, top: Centrum	–	Coarse neighbourhood for the usual destination. Drawn conditional on home_zone from a gravity model balanced against the jobs in each zone, so somebody living in Buitenveldert is far likelier to work on the Zuidas than somebody living in Noord. Equal to home_zone for the retired and for home carers, who have no commute.
Hidden traits — the answer key, never a feature			
truth_taste_vector	2,000 distinct, top: 0.1521;0.0872;0.038…	–	Ten weights summing to 1, one per content category, stored semicolon-separated because CSV has no array type. Decided which items this commuter was shown.
truth_price_sensitivity	0.02 / 0.45 / 0.99	–	How much a price puts them off. Negative coefficient on acceptance.

Column	Values	Null	What it is
truth_coupon_proneness	0.00 / 0.26 / 0.82	–	General appetite for offers. Positive coefficient. Deliberately has no age or gender shift.
truth_detour_tolerance_min	0.50 / 4.75 / 22.00	–	Minutes they will walk out of their way. Enters as a threshold rather than a slope, which is why it is the hardest trait to recover.
truth_novelty_seeking	0.02 / 0.45 / 0.98	–	Probability of being shown a new brand rather than one they already use.
truth_battery_anxiety	0.01 / 0.36 / 0.93	–	Sets a personal disengagement threshold between 20 and 40 percent battery, matching the two curves in the low-battery-anxiety study.
truth_content_appetite	0.24 / 1.01 / 5.21	–	Multiplier on how many items they open per leg.
consent_status	Opted-in, Opted-out, Withdrawn	–	Opted-in, Opted-out or Withdrawn. An opted-out commuter has no behavioural rows anywhere in the dataset, not blank ones.
accessibility_flag	none, reduced_mobility, hearing_impairment, visual_preference, . . .	–	Opt-in. Changes how an offer or notification should be rendered, not whether it is shown.
withdrawal_at	70 distinct, top: 2026-09-01T08:51:01. . .	96%	When the commuter withdrew personalisation consent. Blank for everyone else. No behavioural event exists after this timestamp.
Travel summary			
n_trips	0.00 / 8.00 / 31.00	–	Journeys taken in the window. Scales to roughly 200 a year.
n_legs_total	1.00 / 9.00 / 38.00	1%	Legs across all those trips.
mean_trip_minutes	6.72 / 14.18 / 55.60	1%	Average in-vehicle time per trip.
pct_multimodal	0.00 / 0.00 / 1.00	1%	Share of their trips that crossed modes.
mean_departure_hour	4.67 / 12.40 / 21.50	1%	Average departure hour. Note this averages a twin-peaked distribution, so it lands near midday when almost nobody actually leaves then.
Content summary			
n_views	0.00 / 7.00 / 91.00	–	Content items opened.
mean_dwell_seconds	1 / 50 / 728	19%	Average dwell. Well above the median because of the long tail.
median_dwell_seconds	1 / 29 / 728	19%	The better summary of the two.
bounce_rate	0.00 / 0.08 / 1.00	19%	Share of their views screened out immediately.
top_content_category	10 distinct, top: Local News & Curren. . .	19%	Their single most-viewed category.
share_text	0.00 / 0.45 / 1.00	19%	Share of their views that were text. Rises with age.
share_video	0.00 / 0.55 / 1.00	19%	Share that were video. Falls with age.
Offer summary			
n_offers	0.00 / 4.00 / 28.00	–	Offers shown to them.
n_accepted	9 distinct, top: 0	–	Offers they took.
accept_rate	0.00 / 0.10 / 1.00	17%	The two above divided. Null when no offers were shown, because the rate is undefined rather than zero.
mean_discount_seen	5.00 / 13.75 / 30.00	17%	Average discount they were offered.
mean_detour_seen	0.10 / 6.16 / 33.50	17%	Average detour they were asked to make.
Purchase and points summary			
n_purchases	9 distinct, top: 0	–	Purchases made.
total_basket_eur	0.00 / 4.58 / 35.00	55%	Sum of disclosed basket values.
mean_basket_eur	0.60 / 4.24 / 21.70	62%	Average disclosed basket. Null for commuters whose purchases were all undisclosed.
mean_latency_minutes	0 / 13 / 1,419	55%	Average offer-to-purchase gap. Inflated by the deadline redemptions; prefer the median.
points_balance	0 / 183 / 1,836	–	Spendable points at the end of the window.
n_ledger_entries	10 distinct, top: 1	–	How many ledger rows they have: opening balance, earns, redeems and expiries.
Device summary			
mean_battery	3.00 / 47.39 / 82.00	1%	Average battery across their legs.
pct_offline	0.00 / 0.07 / 1.00	1%	Share of their legs with no connection.
n_reactions	0, 1, 2, 3, 4, 6	–	Likes, saves and dismisses. Median 0 by design.

Column	Values	Null	What it is
commute_minutes	6.00 / 12.00 / 44.00	–	Public transport time from home_zone to work_zone, read straight off the zone matrix. Zero for anyone whose two zones are the same. The mean across commuters is the gravity model's calibration target.

event_stream.csv

One row per event 70,758 rows × 7 columns

Every event from every table in one long chronological file. Use this to replay a single commuter's history in order.

Column	Values	Null	What it is
user_id	2,000 distinct, top: u_00368	–	The commuter this row belongs to. Joins to combined_users and to every other file.
The event			
occurred_at	29,918 distinct, top: 2026-08-24 00:00:00	–	When the event happened. The file is sorted by this column.
event_type	content_view, leg_board, trip_start, offer_shown, points_entry,...	–	Which kind of event this row is: trip_start, leg_board, content_view, content_reaction, offer_shown, purchase, redemption or points_entry.
event_id	70,758 distinct, top: pl_0000001	–	The primary key of whichever table this event came from.
subject	125 distinct, top: Metro	–	The thing the event was about. Meaning depends on event_type: a line for a leg, a category for a view, a merchant category for an offer.
value	-943 / 10 / 1,836	4%	The event's single number. Again depends on type: delay minutes, dwell seconds, discount percent, basket euros, points delta.
detail	1,454 distinct, top: Commute	10%	A short qualifier: the trip purpose, the crowding level, the completion state, the offer outcome, the confirmation status.